A function f defined on an open interval D is said to be *real analytic* on D if for any $x_0 \in D$ the Taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

converges to $f(x)$ for all x in an open interval containing x_0. We will now see that there exist non-analytic functions.

Consider the function

$$g(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ c, & x = 0. \end{cases}$$

In the following we will compute the coefficients of the Maclaurin series $\sum_{n=0}^{\infty} x^n$ for $g(x)$, and show that $g(x)$ is not equal to the sum of its Maclaurin series.

1.) For which value of c will $g(x)$ be continuous at $x = 0$?

2.) Find c_0 in the Maclaurin series for $g(x)$.

3.) Compute $g'(x)$ for $x \neq 0$.

4.) Compute $g'(0)$. Since the function is piecewise defined, you will need to use the definition of the derivative $g'(0) = \lim_{h\to 0} \frac{g(0+h)-g(0)}{h}$. You may freely use the fact that $e^{-x} \to 0$ as $x \to \infty$ faster than any polynomial in x; i.e., if $p(x)$ is any polynomial in x, then $\lim_{x\to 0} \frac{e^{-x}}{p(x)} = 0$.

5.) Find c_1 in the Maclaurin series for $g(x)$.

6.) Write $g'(x)$ as a piecewise function.

7.) Compute $g''(x)$ for $x \neq 0$.

8.) Compute $g''(0)$ following the same strategy you used to compute $g'(0)$.

9.) Find c_2 in the Maclaurin series for $g(x)$.

10.) Do you see a pattern in your previous calculations? Use this pattern to find c_n in the Maclaurin series for $g(x)$.

11.) Find the sum of the Maclaurin series for $g(x)$.

12.) For which values of x is $g(x)$ equal to the sum of its Maclaurin series? Explain.